

Akshat

Data Scientist | Python, Machine Learning, NLP, SQL

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Summary

AI/ML graduate with hands-on experience building and evaluating machine learning models from **transformer-based NLP classifiers** and **classical computer vision** to **retrieval-augmented generation** and a **language model built from scratch**. Comfortable working across the full pipeline: data collection, cleaning, model training, and evaluation.

Professional Experience

Data Analyst Intern, Linengrass

08/2025 – 09/2025

Scraped and cleaned Bangalore hotel data (name, address, amenities, pricing) from multiple sites — 2,200+ of ~2,400 hotels collected, deduplicated, and standardized for analysis in Excel.

Delhi

Projects

Treasury — AI-Powered Thrift Marketplace 🔗

Fullstack resale marketplace with AI sell-assist: **TF-IDF + Logistic Regression** category classifier, **rule-based attribute extraction**, and a **MiniLM-embedding** content recommender (**cosine similarity**, NumPy) for personalized feeds.

Small Language Model (SLM) — Built from Scratch 🔗

Built and pretrained a **30M-parameter decoder-only GPT** (8 layers, 8 heads) with a custom **byte-level BPE tokenizer**, **FlashAttention**, and memmapped data streaming; fine-tuned for Q&A on **Dolly-15k/Alpaca**, fixing a target-alignment bug and adding a free-running generation probe for checkpoint selection.

Hybrid RAG + Calculation Engine 🔗

Local, offline assistant routing queries between a **RAG** pipeline (**MiniLM** embeddings, **cosine similarity**, local LLM via **Ollama**) and a custom-coded **formula engine** for exact numeric computation, avoiding LLM hallucination on precise values.

Manufacturing Defect Detection — Classical CV 🔗

6-class defect classifier (1,344 industrial images) using hand-engineered features (**HOG**, **LBP**, **GLCM**) with an **SVM**, achieving **94.1% test accuracy (97.1% CV)**, outperforming Random Forest; chosen over a CNN for small-data/CPU efficiency.

Fake Job Postings Detection — DistilBERT 🔗

Fine-tuned **DistilBERT** on 17,880 postings (95.2%/4.8% imbalance) with **weighted cross-entropy loss**, achieving **99% accuracy, 0.96 macro F1, 0.989 AUC-ROC**.

Skills

Python (Pandas, NumPy, Scipy, TensorFlow, Keras, scikit-learn, matplotlib) • **SQL** • **NLP/Transformers** • **Computer Vision** (CNN, OpenCV) • **RAG** (sentence transformers, Ollama) • **PowerBi** • **Tableau** • **Excel**

Education

B.Tech in Artificial Intelligence and Machine Learning,
Noida Institute of Engineering and Technology

11/2022 – 05/2026
Greater Noida, UP